

Report Deep Learning

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1 Introduction

Over the past two decades, the space environment has undergone a profound transformation. What was once a domain limited to a few governmental actors has become a highly active industrial and commercial sector. The rapid development of private launch services, the miniaturization of satellites, and the strong demand for global connectivity have led to the massive deployment of spacecraft in low Earth orbit. Today, thousands of active satellites coexist with an ever-growing population of space debris, creating a dense and complex orbital environment.

In this context, accurate orbit prediction has become a critical requirement for the safety and sustainability of space activities. Precise knowledge of satellite trajectories is essential for space surveillance and tracking, collision avoidance, mission operations, and space traffic management. Small prediction errors can lead to inefficient mission planning, increased fuel consumption, loss of communication opportunities, or, in the worst case, catastrophic collisions. As orbital congestion continues to rise, the reliability of prediction tools directly conditions the long-term viability of space operations.

From a physical perspective, satellite motion is influenced by a large number of perturbations. The Earth’s gravitational field is not perfectly spherical and introduces secular effects on orbital elements. For satellites in low Earth orbit, atmospheric drag constitutes the dominant non-gravitational perturbation. Its highly variable nature, driven by solar activity and geomagnetic conditions, makes it particularly difficult to model accurately. Additional effects such as solar radiation pressure and third-body perturbations further contribute to the complexity of orbital dynamics. As a result, even small modelling uncertainties can lead to significant deviations between predicted and true satellite positions over short time scales.

In operational practice, orbital information is commonly distributed through the Two-Line Element (TLE) format. A TLE provides a compact representation of the mean orbital elements of a spacecraft at a given epoch. To compute the position of a satellite at a future time, TLEs must be propagated using an orbital model. The de facto standard for this task is the **SGP4** propagator, which was specifically designed to be compatible with TLE data. This TLE+**SGP4** pipeline is widely adopted due to its simplicity, computational efficiency, and robustness.

However, this operational standard suffers from intrinsic limitations. TLEs represent mean orbital elements rather than instantaneous physical states, and **SGP4** relies on simplified force models. As a consequence, TLE-based orbit prediction is affected by systematic errors that accumulate with time since the last TLE update. For low Earth orbit satellites such as the International Space Station (ISS), prediction errors can reach several kilometres after only a few hours of propagation. While this level of accuracy may be sufficient for visualization or rough tracking, it is inadequate for applications requiring precise orbital knowledge.

Improving the accuracy of TLE-based propagation using purely physical models would require access to detailed force models and high-precision tracking data, which are often unavailable in open operational contexts. This motivates the exploration of data-driven approaches capable of exploiting historical prediction errors to compensate for the limitations of analytical propagation models.

Rather than attempting to replace **SGP4**, this project adopts a hybrid strategy. The objective is to preserve the physical consistency and operational efficiency of the TLE+**SGP4** pipeline while learning a correction model that compensates for its systematic residual errors. Machine learning techniques are particularly well suited for this task, as they can approximate complex nonlinear mappings from large volumes of historical data.

The main objective of this work is therefore to evaluate the ability of deep learning models to improve TLE-based orbit prediction by learning the residual error of **SGP4** with respect to high-precision reference ephemerides. Several neural architectures are investigated, ranging from static feed-forward models to sequential recurrent networks and attention-based Transformer models. The International Space Station is selected as the reference satellite due to its strong sensitivity to atmospheric drag, frequent manoeuvres, and the high availability of accurate reference data.

2 First Learning-Based Approach : Predicting future TLEs

Before introducing the residual correction framework that constitutes the core of this study, a first learning-based approach was explored. This initial strategy aimed at directly predicting future Two-Line Element sets (TLEs) of the International Space Station using a recurrent neural network. Although this approach appeared natural from a time-series perspective, it revealed important physical and methodological limitations. This section presents the formulation of this first attempt, its evaluation, and the reasons that motivated its abandonment.

2.1 Direct Prediction of TLEs

The initial objective was to treat the historical sequence of ISS TLEs as a multivariate temporal signal and to use a Long Short-Term Memory (LSTM) network to predict the next orbital state. In this setting, the network was asked to learn the entire orbital dynamics implicitly, without relying on a physical propagator. Each prediction therefore consisted of a full TLE vector, from which a future orbit could be reconstructed and propagated using SGP4.

This approach directly targets orbit prediction at the level of orbital elements. In theory, such a formulation could bypass SGP4 entirely. In practice, however, it requires the neural network to encode complex physical couplings between all TLE parameters, including temporal evolution, orbital geometry and perturbation effects.

2.2 TLE Dataset and Feature Representation

The dataset used for this first approach consists of approximately 200,000 historical ISS TLEs collected from public sources such as CelesTrak and Space-Track. The raw TLE strings were parsed and converted into a numerical tabular format. Each TLE vector contains temporal information (epoch expressed in several forms), the main orbital elements (inclination, right ascension of the ascending node, eccentricity, argument of perigee, mean anomaly and mean motion), the drag-related parameter B^* , and additional metadata such as the revolution number at epoch.

In the initial configuration, all numerical fields were used both as inputs and as prediction targets. This resulted in a feature dimension of 18 variables. Such a choice maximised the amount of available information but also introduced strong heterogeneity in the physical nature and numerical scale of the variables to be predicted.

2.3 Temporal Learning Setup

The TLEs were first sorted chronologically. Sliding sequences of fixed length $L = 20$ were then constructed so that each input sequence of past TLEs was mapped to the next TLE in time. All features were normalised using Min-Max scaling in order to compensate for the large dispersion of numerical values, from very small eccentricities to very large Unix timestamps.

The dataset was split strictly in time into training, validation and test subsets, with proportions of 70%, 15% and 15%, respectively. This procedure ensures that the most recent orbital states are never seen during training and provides a realistic evaluation of predictive performance.

2.4 LSTM Architecture and Training

The adopted neural architecture is based on a two-layer LSTM with a hidden dimension of 128, followed by a fully connected layer that maps the last hidden state to the 18-dimensional TLE output vector. A dropout rate of 0.1 is applied between recurrent layers to limit overfitting.

The model is trained using the Adam optimiser and a Mean Absolute Error loss. A learning-rate scheduler and early stopping are used to stabilise convergence. In the normalised feature space, the model exhibits a seemingly correct learning behaviour, with a progressive decrease

of both training and validation MAE, as shown in Figure ???. At this stage, the numerical performance appears satisfactory.

2.5 Evaluation in the Physical Space

Once the predictions are transformed back into physical units, the limitations of this approach become apparent. Feature-wise errors reveal very large discrepancies on several critical orbital parameters. In particular, angular variables such as the right ascension of the ascending node, the argument of perigee and the mean anomaly exhibit errors of several tens to hundreds of degrees. Large inconsistencies are also observed on temporal features such as the epoch and the revolution number, as illustrated in Figure ??.

These errors indicate that a good numerical score in the normalised space does not imply physical coherence of the predicted TLEs.

2.6 Physical Validation with SGP4

To assess the real impact of these inconsistencies, a physical validation step was performed using SGP4. For each predicted TLE in the test set, a reference orbit was propagated from the true TLE and compared to the orbit obtained from the predicted TLE over a six-hour horizon. The resulting position errors are shown in Figures ??, ?? and ??.

The observed mean errors reach several thousand kilometres, with maximum values exceeding ten thousand kilometres. Such magnitudes are comparable to the diameter of the Earth and clearly demonstrate that the predicted TLEs are physically unusable for any realistic orbital application.

2.7 Limits of the Direct TLE Prediction Approach

The failure of this first approach is not due to insufficient model capacity but to an incorrect formulation of the learning task. The network is simultaneously asked to predict time-dependent indices, orbital angles affected by periodic discontinuities, and physically coupled orbital parameters, without any explicit constraint enforcing physical consistency.

As a consequence, even small numerical errors in the normalised space translate into dramatic physical inconsistencies after denormalisation and propagation. This makes the direct prediction of full TLE vectors an ill-posed problem for data-driven learning.

2.8 Motivation for a Residual Correction Strategy

These observations led to a fundamental reformulation of the problem. Instead of trying to predict TLEs directly, the strategy was shifted toward learning a correction of the SGP4 propagation error itself. By retaining SGP4 as the physical backbone of the prediction pipeline and by learning only the residual error in Cartesian space, the learning task becomes both physically meaningful and numerically well conditioned.

This residual learning framework constitutes the foundation of the rest of the study and is introduced in the following section.

3 Second Approach : residual fonction learning

This section presents the dataset construction process, the evaluation methodology, and the learning strategies adopted after the reformulation of the problem as a residual correction task. The objective is to describe clearly how the data were generated, how the models were trained and evaluated, and how the successive model families were introduced in a coherent experimental framework.

3.1 Datasets and Baseline SGP4 Error

Two main datasets were used in this study. The first one concerns the International Space Station (ISS) and contains 17,627 samples constructed from approximately 200 consecutive TLEs. The second dataset corresponds to the Hubble Space Telescope (HST) and contains 48,239 samples over a longer temporal span. These two satellites were selected because they are well tracked, operate in different dynamical regimes, and provide reliable reference ephemerides.

Each sample contains the SGP4-predicted Cartesian position $(x, y, z)_{\text{SGP4}}$, the corresponding Horizons reference position $(x, y, z)_{\text{Horizons}}$, and the residual vector (dx, dy, dz) . The Euclidean residual norm is also stored and used as the principal evaluation quantity.

Before introducing any machine learning correction, the intrinsic error of SGP4 is quantified. For the ISS dataset, the baseline three-dimensional RMSE is approximately 17 km, with a median error close to 4 km. For the HST dataset, the baseline RMSE is around 4.4 km. These values confirm that SGP4 exhibits a strong systematic drift with respect to high-precision ephemerides, especially in low Earth orbit.

3.2 Construction of the Residual Dataset

The dataset construction follows a strict and fully reproducible pipeline. For each satellite, the associated NORAD identifier and Horizons body identifier are first retrieved. A history of approximately 200 TLEs is then collected from public catalogues. For each pair of consecutive TLEs, the first one is propagated using SGP4 up to the epoch of the next TLE, with a fixed time step.

At each propagation timestamp, the Horizons system is queried to obtain the corresponding high-precision Cartesian position. The residual vector is then computed as the difference between the Horizons and SGP4 positions. This procedure produces a dense time series of residual errors between two successive TLE updates.

The learning objective is to predict this residual vector directly and to construct the corrected position as:

$$\mathbf{r}_{\text{corr}}(t) = \mathbf{r}_{\text{SGP4}}(t) + \widehat{\Delta \mathbf{r}}(t).$$

This formulation ensures that the physical structure of SGP4 is preserved, while systematic model errors are compensated by the neural network.

3.3 Preprocessing and Feature Representation

Each input sample gathers three types of information: temporal features (expressed in Unix seconds and in time since the last TLE), raw TLE orbital parameters, and the SGP4-predicted position and velocity components. The target vector corresponds exclusively to the residual components (dx, dy, dz) .

All input features are normalised using Z-score normalisation in order to stabilise training and to avoid scale domination effects. The target residual vector is also normalised, which was found to be essential for the stability of Transformer-based models. Sliding temporal windows of fixed length (between 50 and 128 time steps) are constructed to provide the models with temporal context.

A strict chronological split is applied. In addition, a split by TLE blocks is used so that all samples belonging to the same TLE interval are assigned to the same subset. This guarantees the absence of any temporal leakage between the training, validation and test sets.

3.4 Evaluation Methodology

All models are evaluated exclusively on test data that are strictly unseen during training. The primary metric is the three-dimensional RMSE on the residual vector (dx, dy, dz) , as well as the Euclidean position error after correction. All corrected Cartesian trajectories $(x, y, z)_{\text{corr}}$ are exported and compared directly to the Horizons reference.

For models involving long temporal windows, an additional extra-test dataset is used to evaluate out-of-domain generalisation. This strategy ensures that the reported results reflect not only interpolation capability but also true extrapolation performance.

3.5 Model Families and Learning Strategies

Several families of neural architectures were evaluated progressively. The first baseline corresponds to Multilayer Perceptrons (MLPs), applied to the instantaneous SGP4 state at the end of each temporal window. These models offer fast training and simple optimisation but do not explicitly model temporal dependencies.

Sequential architectures were then introduced through Long Short-Term Memory networks and Gated Recurrent Units. These models process the full temporal window and aim at learning the dynamic evolution of the residual error. Various depths and hidden dimensions were explored, along with dropout and learning-rate schedulers for stability.

Finally, a Transformer encoder architecture was implemented. This model relies on multi-head self-attention to capture both short- and long-range temporal dependencies. It includes input normalisation, causal attention masking, AdamW optimisation, learning-rate warm-up, and explicit normalisation of the target residual vector. This architecture represents the most advanced model used in the project.

3.6 Training Protocol and Monitoring

All models are trained using a consistent protocol in order to ensure fair comparison. Optimisation is performed using Adam or AdamW, with a batch size of 32. Early stopping and learning-rate scheduling are systematically applied to avoid overfitting and to stabilise convergence.

TensorBoard is used throughout the training process to monitor training, validation and test losses, learning-rate evolution, and global optimisation behaviour. This monitoring proved essential to identify unstable configurations, poorly conditioned learning setups and generalisation issues.

This experimental framework provides a coherent and controlled basis for the comparative analysis presented in the next section.

4 Results and Analysis

This section presents the analysis of the different models trained to correct the SGP4 orbital prediction error. The interpretation is based on the temporal evolution of the position error and on the training metrics observed during optimization. The corresponding figures are provided in the appendix and are referenced throughout this section to support the discussion.

4.1 Baseline Behaviour of SGP4

Before any correction, the raw SGP4 propagation exhibits a characteristic drift behaviour. The position error increases progressively with time since the TLE epoch until a new TLE update becomes available. This pattern is visible in all temporal plots and reflects the accumulation of deterministic modelling errors, mainly due to atmospheric drag and mean motion estimation. This baseline behaviour provides the reference level against which all learning-based corrections are evaluated.

4.2 Behaviour of LSTM Models

Several LSTM configurations were evaluated with different depths and hidden dimensions. The corresponding error evolution curves are shown in Figures , 1, 2, 3 and 4.

Globally, the LSTM models exhibit a more structured behaviour than the static models, but their correction remains limited. In several cases, the corrected error follows the general trend of the SGP4 baseline without significantly reducing its magnitude. Some configurations show noticeable instabilities, especially at TLE boundaries, where abrupt variations appear in the corrected error. This indicates that the recurrent networks struggle to robustly capture the deterministic structure of the SGP4 residual.

The corresponding training curves (Figures 11, 12, 13 and 14) confirm this observation. While the training loss decreases, the validation and test losses remain high and show limited improvement. This behaviour reveals a restricted generalisation capability and a strong sensitivity to architectural depth.

Overall, LSTMs provide only a modest improvement over the raw SGP4 propagation and cannot be considered as a reliable correction solution in this context.

4.3 Behaviour of MLP Models

Several MLP architectures were also evaluated, with different depths and learning rates. The corresponding temporal error plots are shown in Figures 5, 6, 7 and 8.

The MLP models exhibit highly unstable correction behaviours. The corrected error often oscillates strongly and frequently exceeds the baseline SGP4 prediction. In many intervals, the MLPs amplify the original modelling error instead of reducing it. This behaviour reflects the inability of static architectures to correctly handle the temporal structure of orbital dynamics.

The corresponding training metrics (Figures 15, 17, and 18) reveal clear signs of overfitting. Although the training loss decreases, the validation and test losses remain very large, indicating that the learned mapping does not generalise to unseen data.

These results confirm that MLP models, despite their numerical efficiency, are not adapted to the correction of SGP4 orbital propagation.

4.4 Behaviour of the Transformer Model

The Transformer-based model exhibits a radically different behaviour. The temporal evolution of the corrected error, shown in Figures 9 and 10, indicates that the corrected error remains consistently below the baseline SGP4 error over the entire propagation window. The correction

is smooth, stable, and physically coherent, including near TLE boundaries where second-order effects are usually the strongest.

The corresponding training curves in Figure 19 show a progressive and stable convergence on all datasets. Unlike the other model families, the Transformer does not exhibit severe overfitting and maintains a good balance between training and test performance. This behaviour indicates a strong generalisation capability.

The attention mechanism allows the model to exploit both local and long-range temporal dependencies, which is essential for capturing the structured nature of SGP4 residuals.

4.5 Global Comparison of the Models

The comparison of all models leads to a clear performance hierarchy. The MLP models are the least reliable and introduce strong instability. The LSTM models improve slightly upon static models but remain limited in robustness and generalisation. The Transformer model clearly outperforms all other architectures in terms of stability, error reduction, and physical coherence.

This ranking is consistent across both the temporal error evolution and the training metrics, and is fully supported by the figures provided in the appendix.

4.6 Physical Interpretation of the Corrections

The improvements provided by the Transformer mainly affect the along-track component of the orbital error, which is known to dominate SGP4 inaccuracies. This confirms that the model effectively captures systematic biases related to mean motion and atmospheric drag modelling. Radial and cross-track components are also slightly improved, but to a lesser extent.

These observations confirm that the residual correction strategy is physically consistent and that the learned correction reflects real limitations of the SGP4 model rather than numerical artefacts.

4.7 Synthesis

The analysis of the experimental results leads to the following conclusions. Static models are not suitable for correcting orbital propagation errors. Recurrent architectures provide only limited improvement and remain unstable for several configurations. The Transformer architecture is the only model that achieves a stable, consistent, and physically meaningful correction of SGP4 predictions.

This result validates the relevance of attention-based models for the correction of deterministic orbital propagation errors and justifies their use in the perspective of operational SGP4 enhancement.

5 Conclusion and Perspectives

This project explored the use of deep learning to correct the deterministic errors of the SGP4 orbital propagator using TLE data. The work followed a clear and progressive methodology, starting from static MLP models, moving toward recurrent architectures, and finally reaching an attention-based Transformer model. This evolution was guided both by physical validation through orbital error analysis and by rigorous monitoring of training behaviour.

The experimental results show a clear hierarchy between the tested models. MLP architectures proved inadequate for this task and frequently introduced instabilities. LSTM models provided slightly improved structure but remained limited in robustness and generalisation. The Transformer clearly stands out as the only model capable of producing a stable, coherent, and physically meaningful correction of SGP4 predictions. Its ability to exploit long-range temporal dependencies allows it to capture the structured nature of SGP4 residual errors more effectively than the other architectures.

Several concrete improvements now emerge as natural next steps. The integration of explicit positional encoding or Time2Vec features could further strengthen temporal representation. The use of more robust loss functions such as MAE or Huber loss would reduce sensitivity to outliers. A stricter validation strategy based on cross-validation by TLE blocks would further reinforce the reliability of the results. Feature ablation studies would help identify the most informative inputs, and multi-satellite training would allow the assessment of model generalisation beyond the ISS case.

Overall, this work demonstrates that Transformer-based models constitute a reliable and scalable solution for improving TLE-based orbit propagation. It opens the way toward robust learning-based correction layers that can be integrated into operational SGP4 pipelines for enhanced orbit prediction accuracy.

A Appendices

This appendix gathers the main figures used to support the analysis presented in the report. They include error evolution over time for each model and the corresponding training metrics.

Error evolution plots

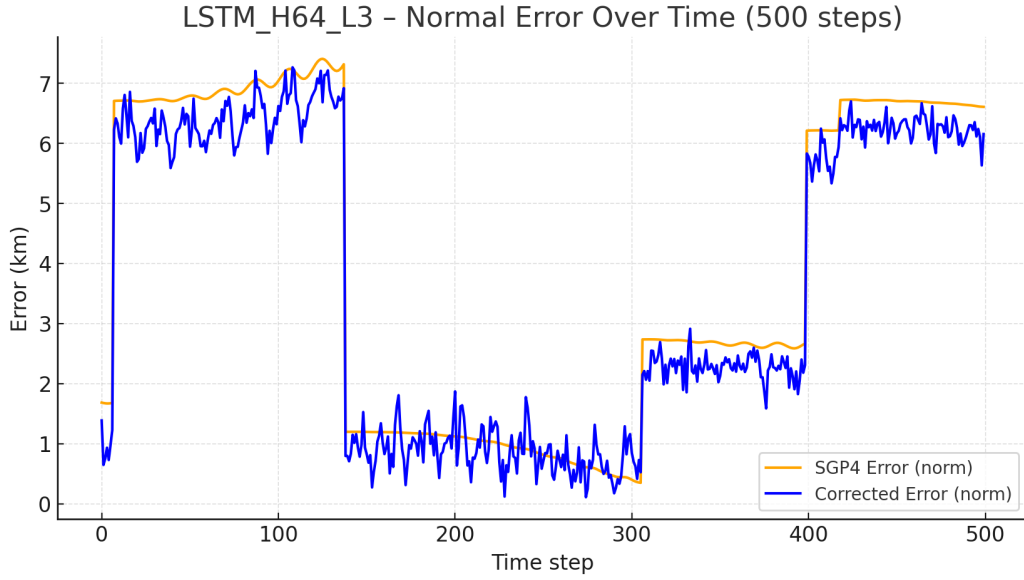


Figure 1: Error over time for LSTM model (H64, 3 layers).

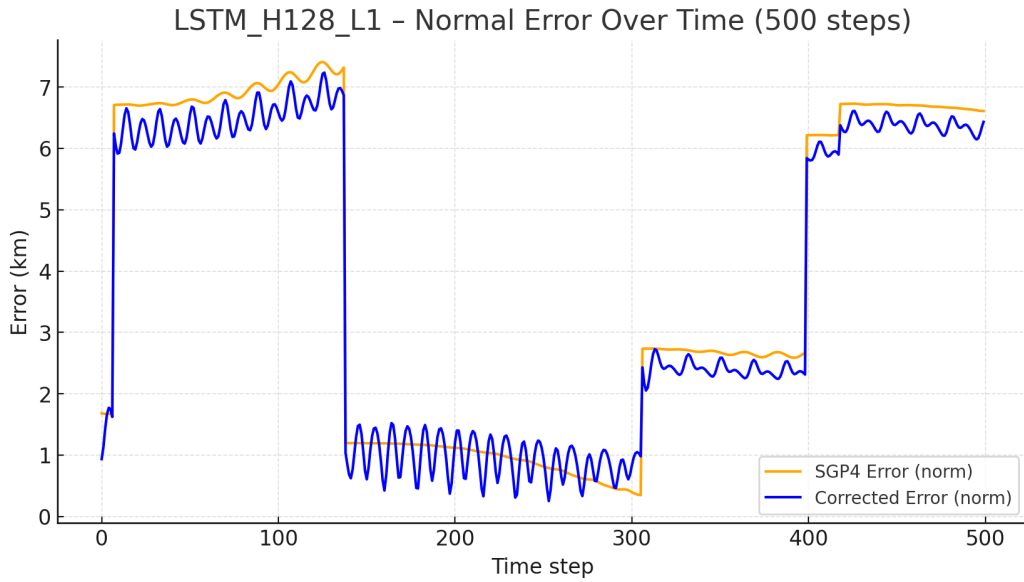


Figure 2: Error over time for LSTM model (H128, 1 layer).

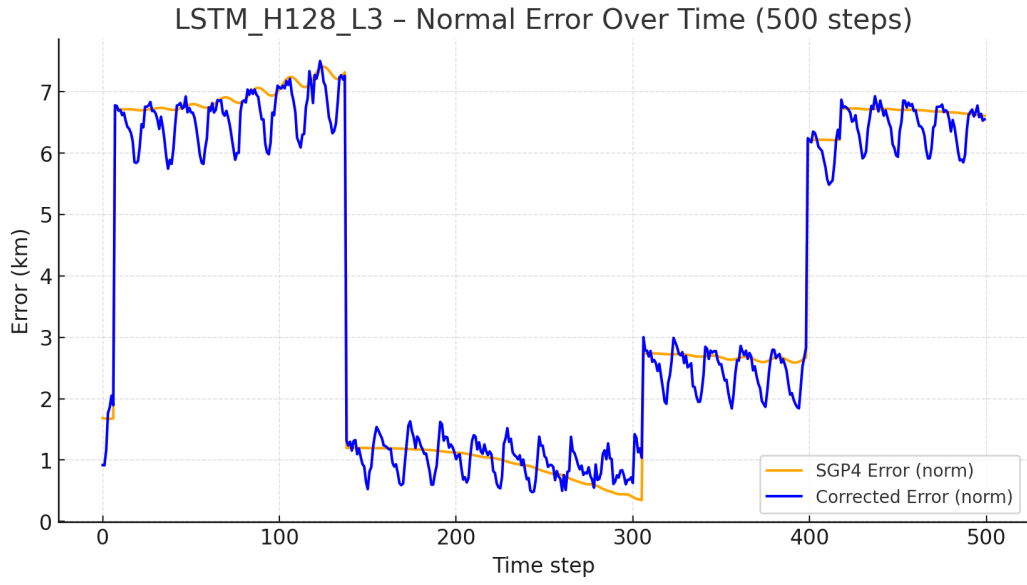


Figure 3: Error over time for LSTM model (H128, 3 layers).

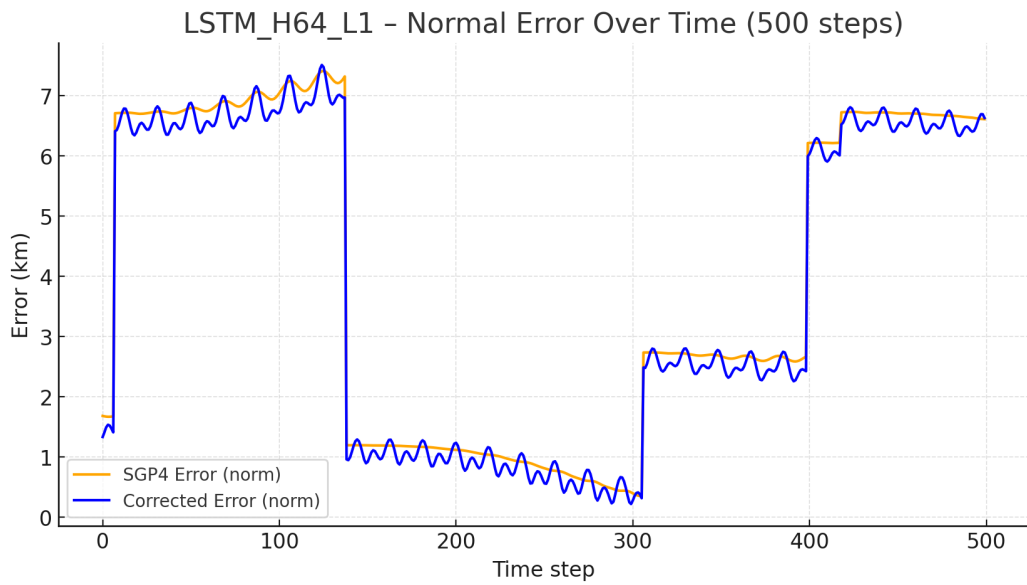


Figure 4: Error over time for LSTM model (H64, 1 layer).

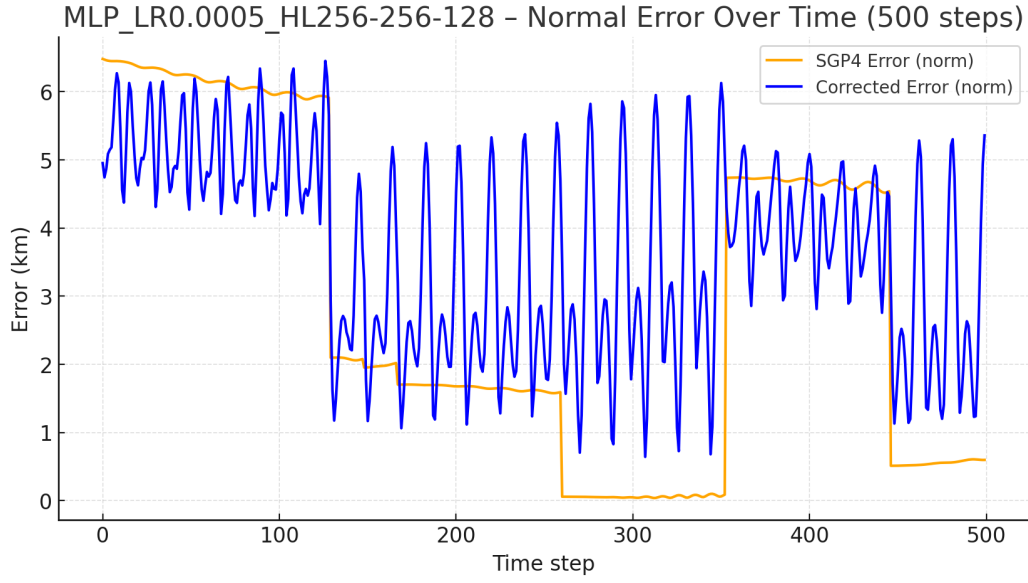


Figure 5: Error over time for MLP model (HL256–256–128, $\text{LR}=5 \times 10^{-4}$).

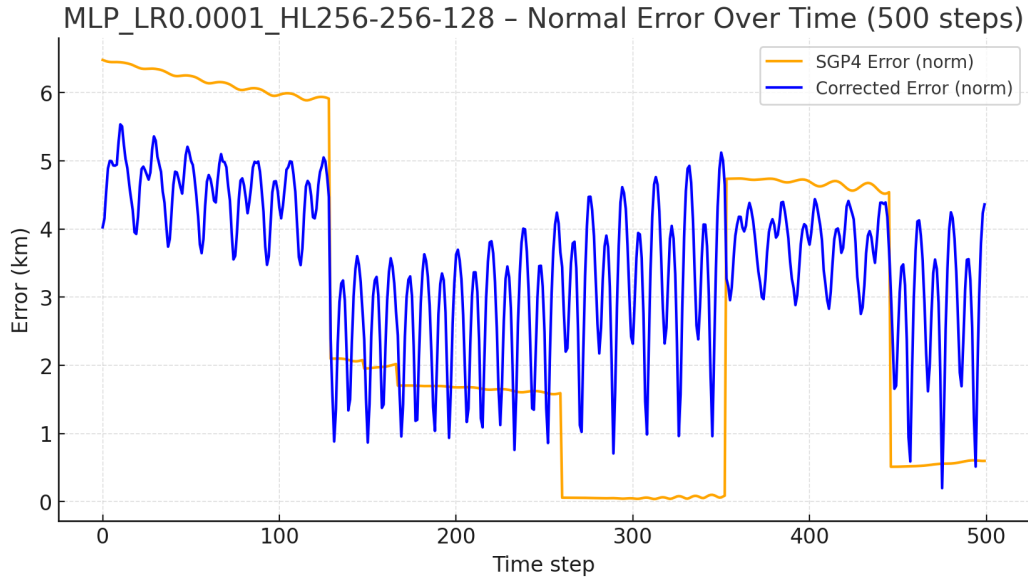


Figure 6: Error over time for MLP model (HL256–256–128, $\text{LR}=10^{-4}$).

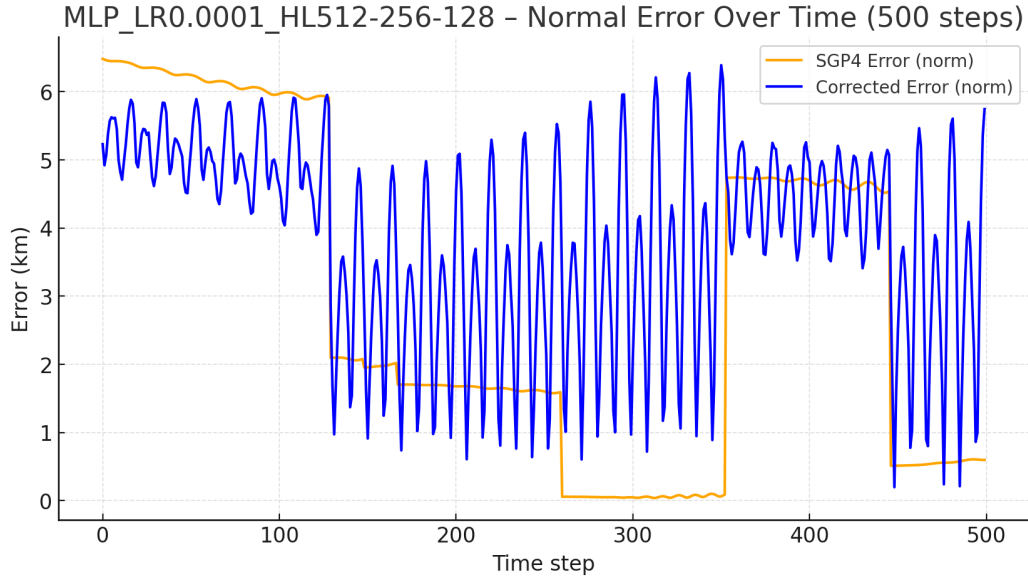


Figure 7: Error over time for MLP model (HL512-256-128, $LR=10^{-4}$).

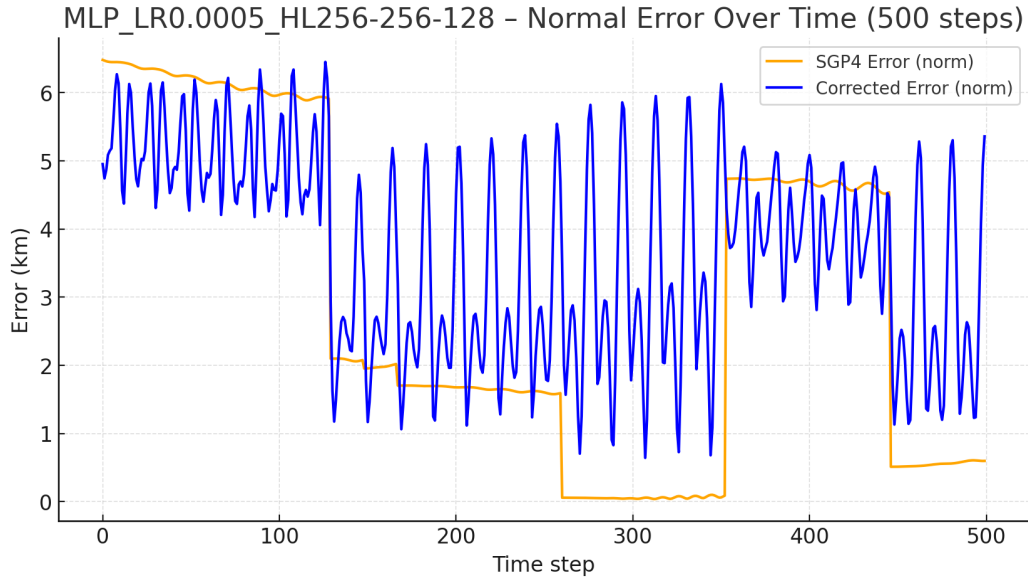


Figure 8: Error over time for MLP model (HL256-256-128, $LR=10^{-3}$).

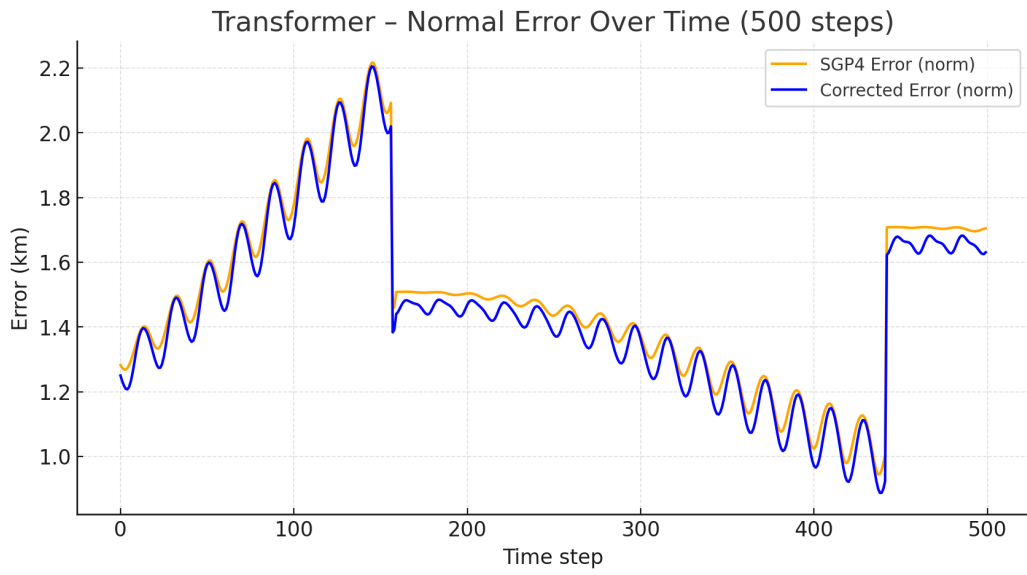


Figure 9: Error over time for the Transformer model.

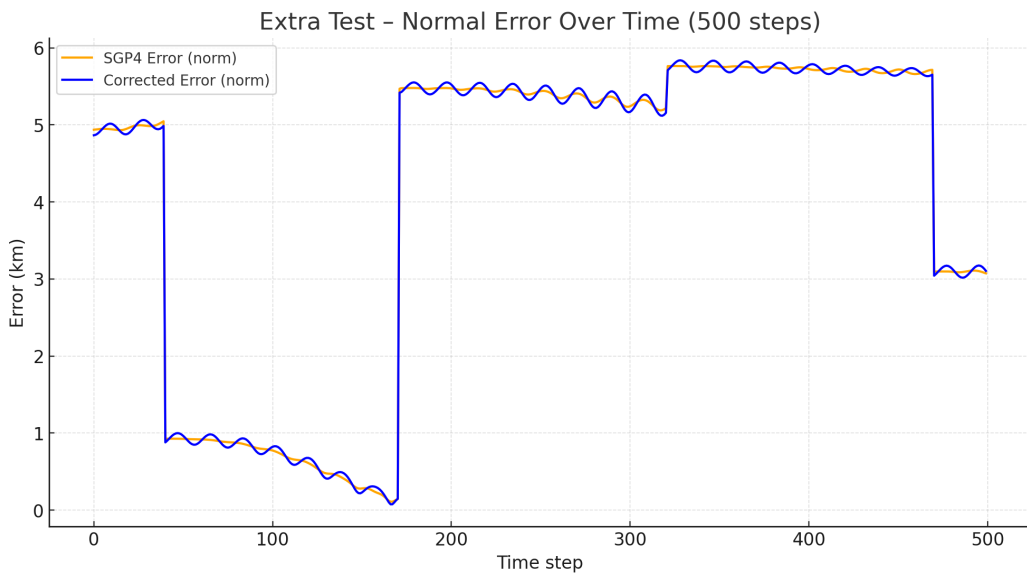


Figure 10: Error over time for the Transformer model on ISS data.

Training metrics

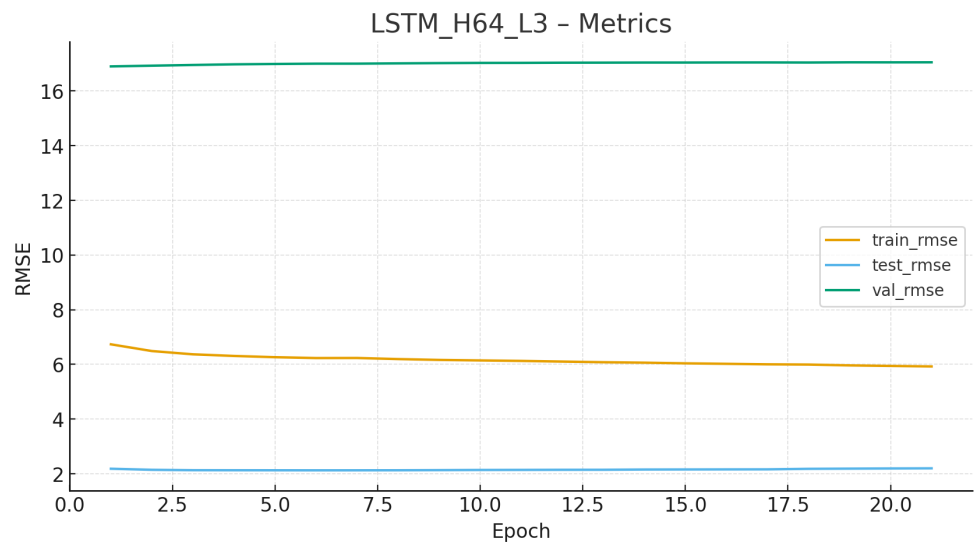


Figure 11: Training, validation and test metrics for LSTM (H64, 1 layer).

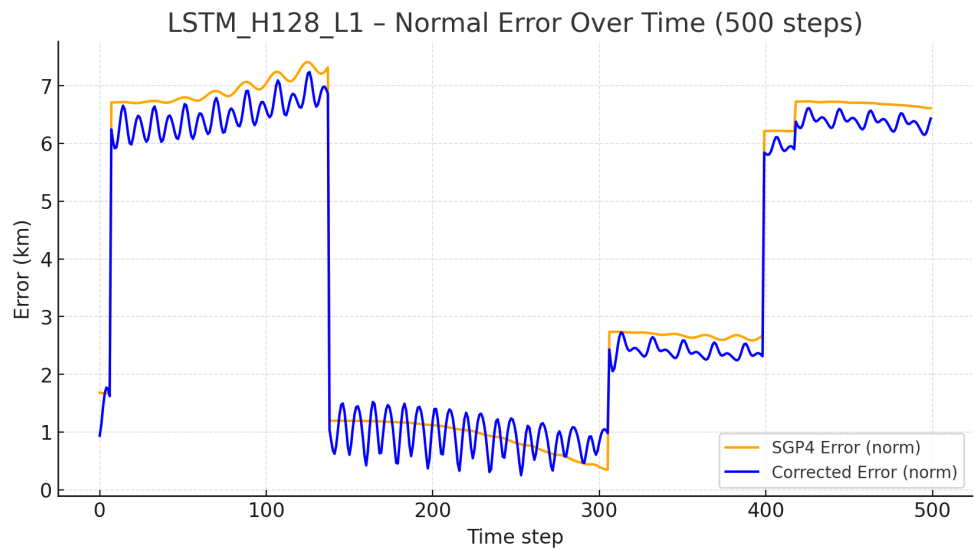


Figure 12: Training, validation and test metrics for LSTM (H64, 3 layers).

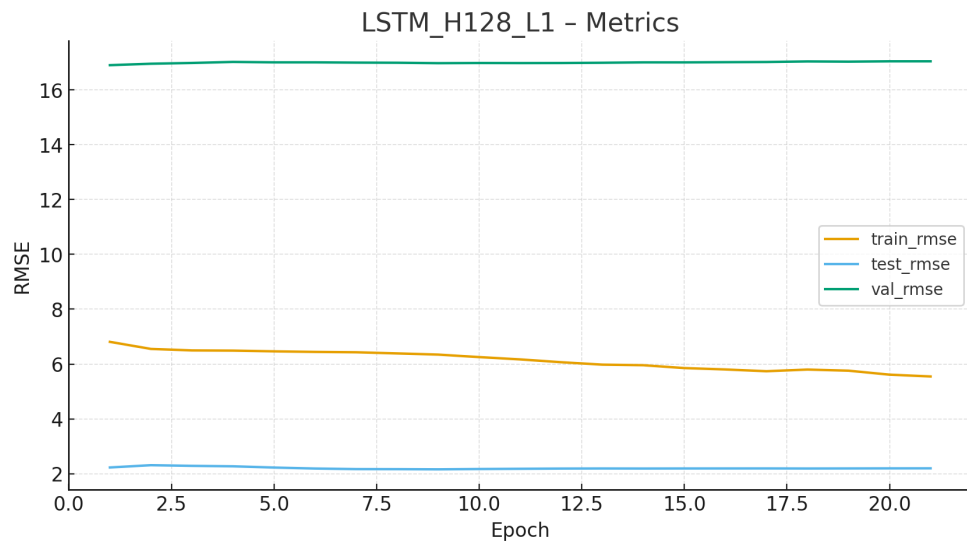


Figure 13: Training, validation and test metrics for LSTM (H128, 1 layer).

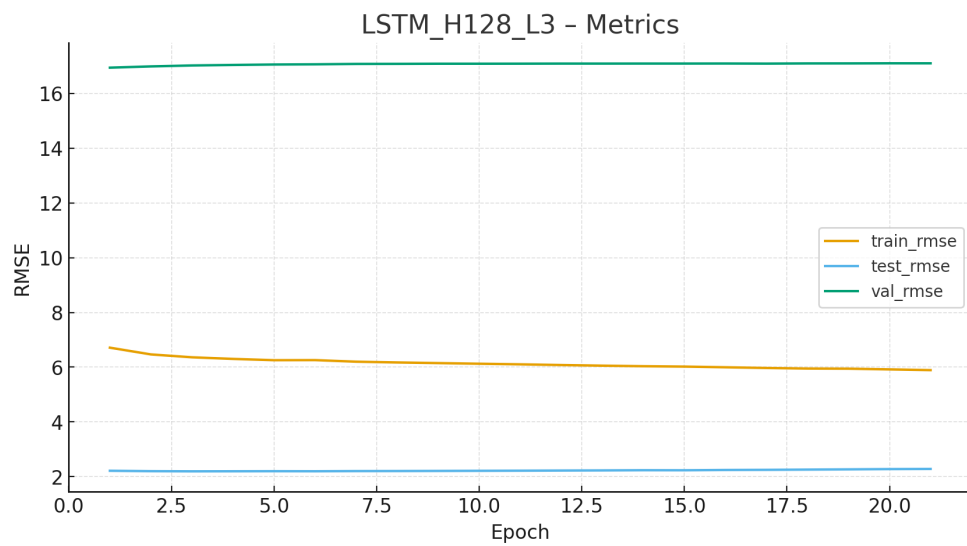


Figure 14: Training, validation and test metrics for LSTM (H128, 3 layers).

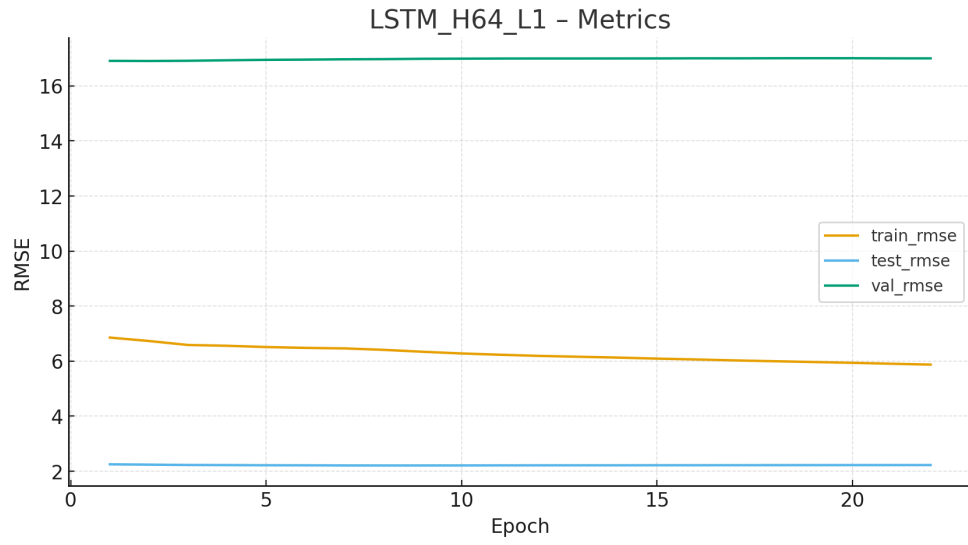


Figure 15: Training, validation and test metrics for MLP (HL256–256–128, $\text{LR}=5 \times 10^{-4}$).

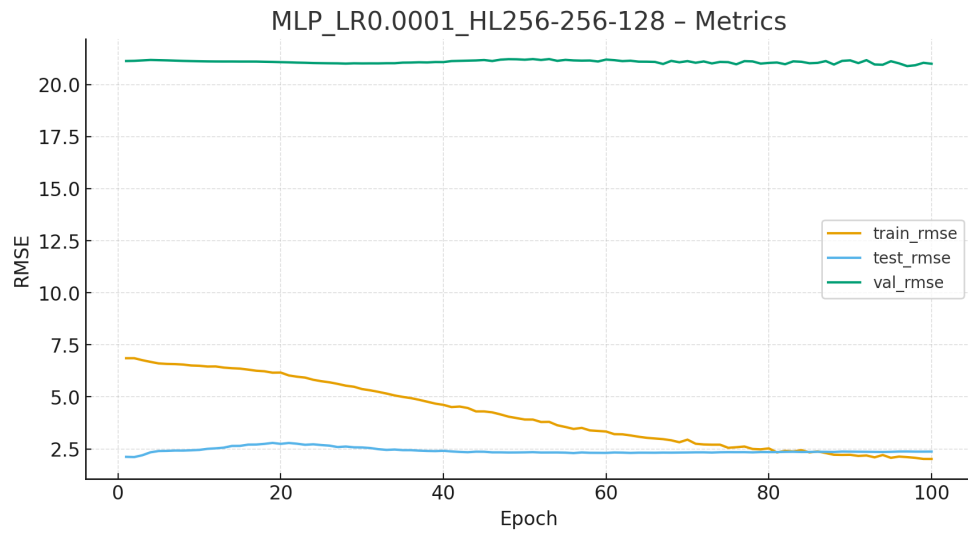


Figure 16: Training, validation and test metrics for MLP (HL256–256–128, $\text{LR}=10^{-4}$).

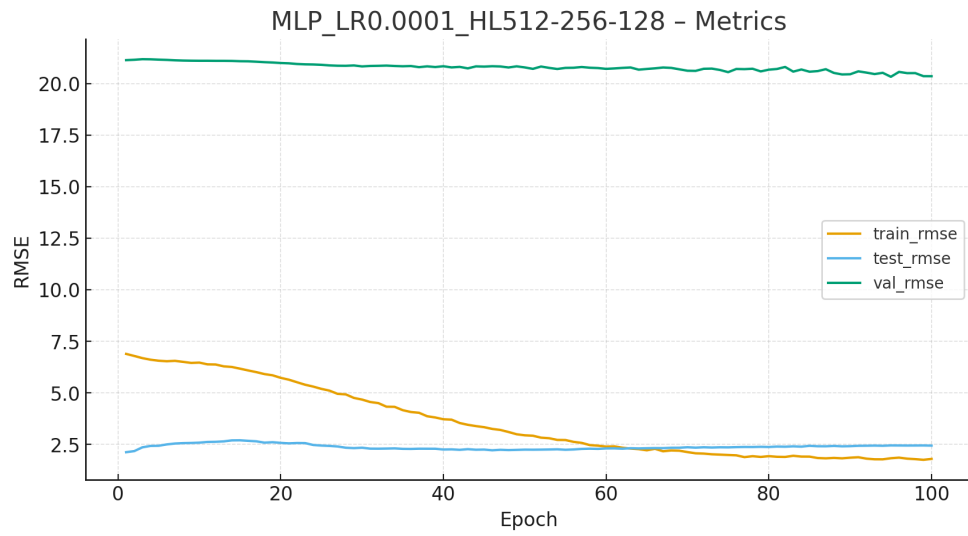


Figure 17: Training, validation and test metrics for MLP (HL256-256-128, $LR=10^{-4}$, alternative run).

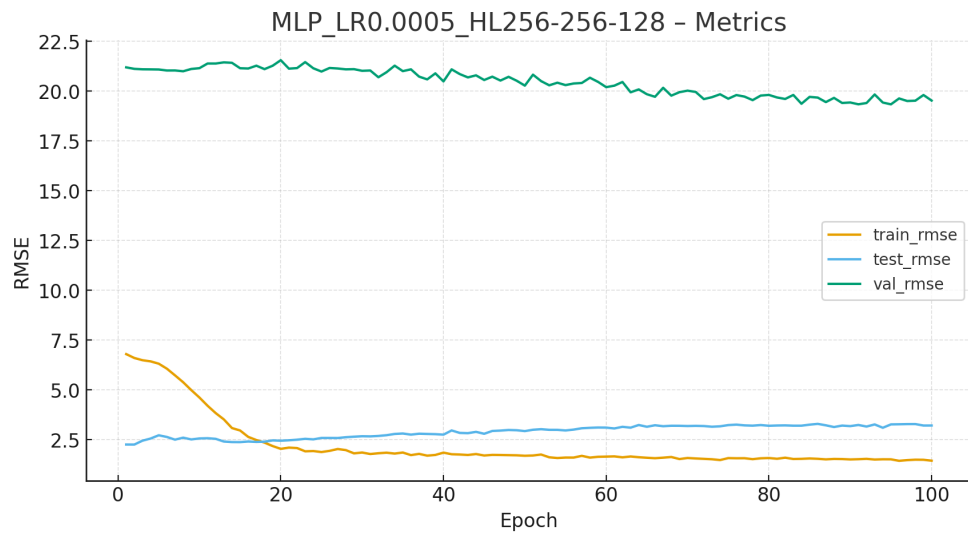


Figure 18: Training, validation and test metrics for MLP (HL512-256-128, $LR=10^{-4}$).

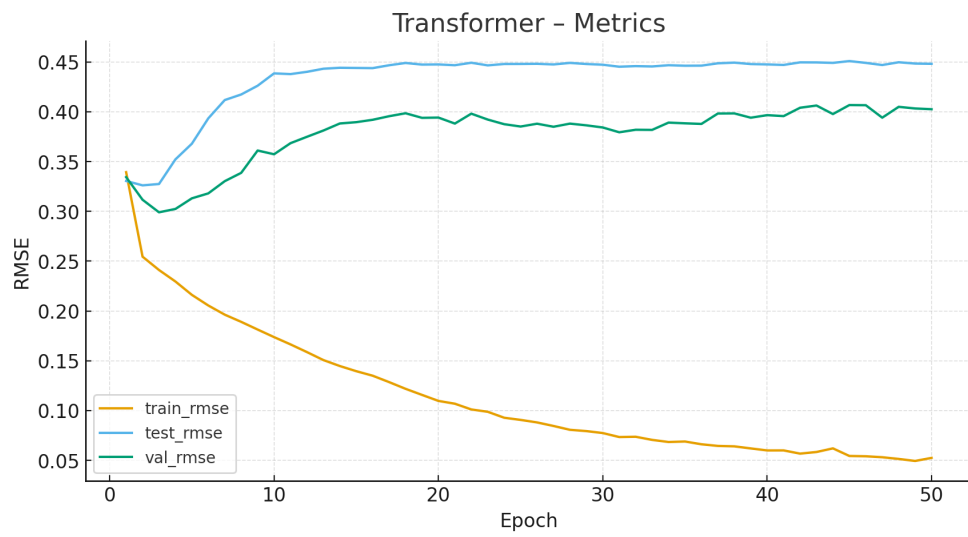


Figure 19: Training, validation and test metrics for the Transformer model.